MINING DEALER BEST PRACTICE

Tool for Installing 3412 & 3408 Engines Injection Pumps

Maintenance and Repair | Application | Component Life Management | **Component Renewal (CRC)** | MARC Management

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1.0 Introduction

The 3412 and 3408 engines use the direct injection system, and the HEUI system for the fuel supply to the injectors. The direct injection system uses a large pump that is located at the top of the engine block, where it fits in the engine by using a seal to prevent oil leakage. In the installation process of the pump; the levers or slings are used (Figure 1); this method puts the technician, the seal and some other engine components at risk.

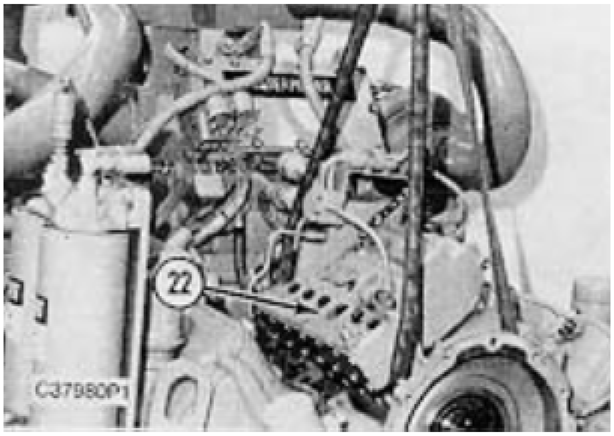

Figure 1. Tools for pump installation using slings.


Figure 2. Tools for pump installation using lever.

A tool that ensures a safe and fast installation process for the technicians and the machine is now presented.

2.0 Best Practice Description

When fitting into the engine, the injection pumps have a seal that is commonly damaged during the installation process, this is mainly due to the method used which is not precise enough to guide the bomb back to its final position into the engine.

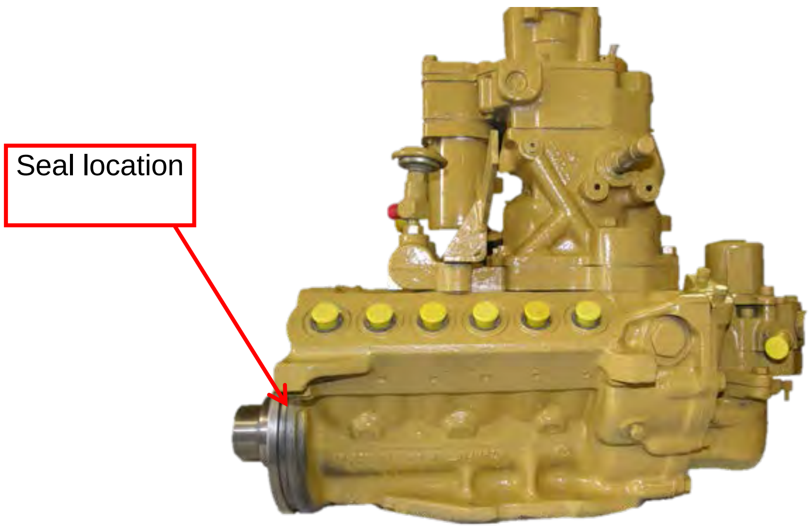

Figure 3. Direct Injection System Pump

The designed tool (Figure 4) consists of 6 parts, which allow it to have the mobility needed for its application. The pump is fixed by the camshaft end thru 4 bolts to coupling "A" of the tool; (See Figure 4). This coupling is connected to a main bolt "B" and this one spins on the "C" base that is held to the engine housing. The "C" base converts the bolt rotational movement in the linear movement of the pump by placing it concentrically and take it to a final position, smoothly and precisely.

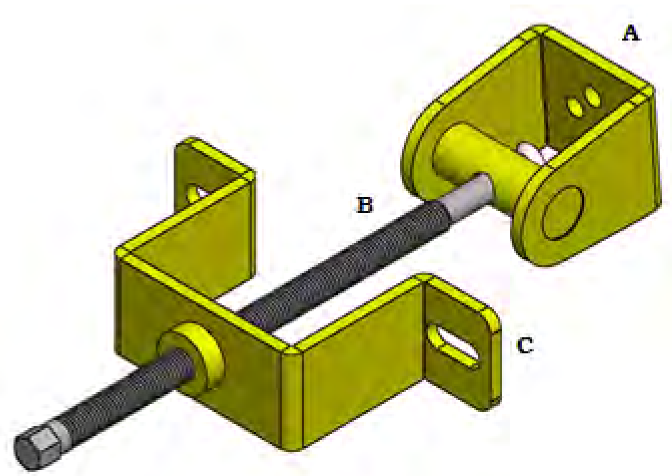

Figure 4. Tool for installing the injection pump

3.0 Implementation Steps

For the installation of the pump, follow the following steps (See Figure 5).

1. Place the pump on top of the engine using a lifting device (The weight of the fuel injection pump housing and governor is 65 kg). Then, place the tool coupling to the pump camshaft end by using 2 bolts (9M-6590).
2. Install the tool support on the housing timing gear using two bolts (0S-1594) and their washers (5M-2894).
3. Insert the main tool bolt into the base and turn to be long enough so that the tip of the bolt passes through the pin in the tool coupling. Secure the bolt to the coupling using a hex bolt.
4. Turn the main bolt using a wrench or pneumatic tool to bring the pump nearer to its final position.
5. Continue with the location of other accessories of the pump according to the SIS manual on injection pumps.

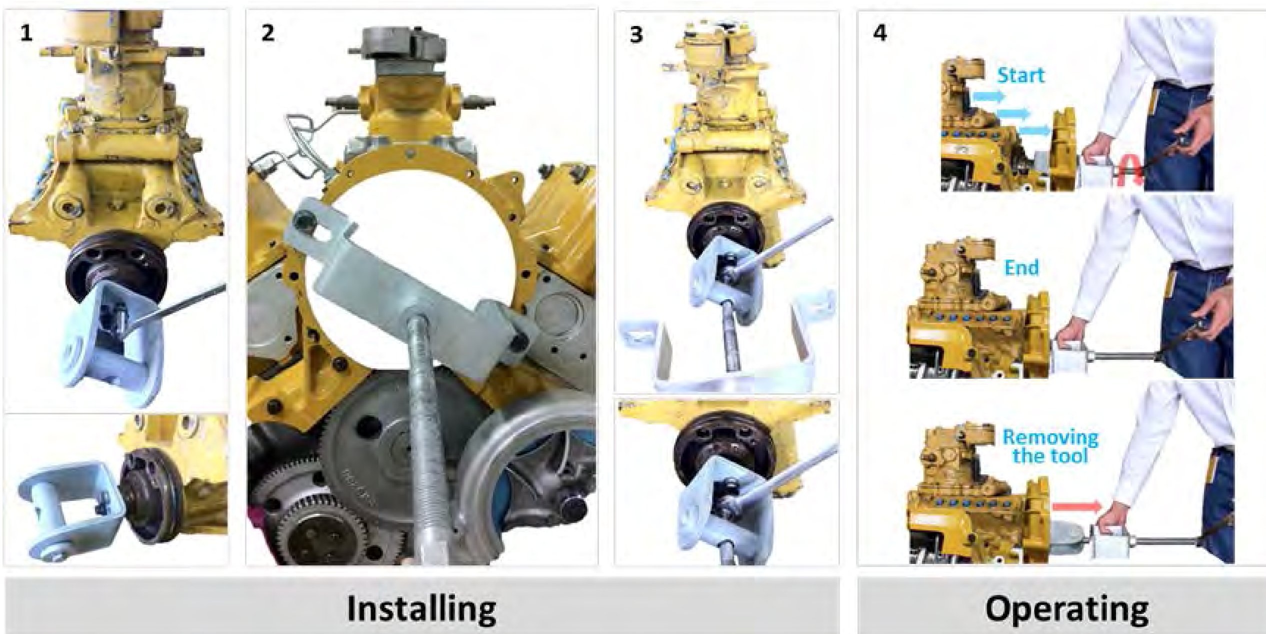

Figure 5. Implementation steps

4.0 Benefits

- It facilitates the task of installing the injection pump on 3412 engines during its assembly process
- It reduces the risk of accidents for technicians when the lever is not used.
- It protects the engine components from bumps, scratches or dents and minimizes the risk of damaging the seal (PN: 6V-4314, value: USD $ 28).

5.0 Resources Required

The manufacturing cost of the tool is USD $210, the additional attachments for the installation are listed in Table 1. In the document *Tool Drawing* (Injection pump).pdf the dimensions can be seen (See attachments).

Table 1. Hardware Required		
Part Number	Description	Qty
0S-1594	3/8-16X1 inch Bolt	2
9M-6590	1/2-13 x 1 inch Bolt	2
9L-9132	3/8 inch Washer	2

6.0 Supporting Attachments / References

- SENR2095-01: Fuel Injection Pump Housing And Governor
- Tool Drawing (Injection pump).pdf

7.0 Related Best Practices

None

8.0 Acknowledgements

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